

EMPLOYEE COST

1. Labour Turnover

It is the rate of change in labour force of an organisation. It can be calculated by following ways:

- (A) Separation Method = $\frac{\text{Number of Separation}}{\text{Average No. of Workers}} \times 100$
- (B) Replacement Method = $\frac{\text{Number of Replacement}}{\text{Average No. of Workers}} \times 100$
- (C) New Recruitment Method = $\frac{\text{Number of Expansion}}{\text{Average No. of Workers}} \times 100$
- (D) Flux Method (with replacement) = $\frac{(\text{No. of Separation} + \text{No. of accessions})}{\text{Average No. of Workers}} \times 100$
- (E) Flux Method (without replacement) = $\frac{(\text{No. of Separation} + \text{No. of replacement})}{\text{Average No. of Workers}} \times 100$
- (F) Average worker = $\frac{\text{Opening} + \text{Closing}}{2}$
- (G) Equivalent Annual LTR = $\frac{\text{Labour Turnover Rate}}{\text{No. of days in the period}} \times 365$

2. Statement of Profit foregone due to labour turnover

Particulars	Amount
Contribution lost due to loss of productive hours → Cont. Per Hour × Prod. Hrs. lost	
Additional rectification cost	
Settlement cost due to leaving	
Recruitment cost	
Training cost	
Profit foregone	

Question – 1

The cost accountant of SK Ltd. has computed labour turnover rates for the quarter ending 31st March as 10%, 5% and 3% respectively under 'Flux Method', 'Replacement Method' and 'Separation Method'. If the number of workers replaced during the quarter is 30, find out the number of:

- (a) workers recruited and joined; and
- (b) workers left and discharged
- (c) equivalent employee turnover rates for the year
- (d) number of workers at the beginning of the quarter

$$\text{Op.} + \text{Join} - \text{Sep.} = \text{Cl.}$$

Solution Op. + 42 - 18 = Cl. ⇒ Op. + 24 = Cl.

$$\frac{5}{100} = \frac{30}{\text{Avg.}} \Rightarrow \text{Avg.} = 600$$

$$\frac{3}{100} = \frac{\text{Sep.}}{\text{Avg.}} \Rightarrow \text{Sep.} = 600 \times 3\% = 18$$

$$\frac{10}{100} = \frac{\text{Sep.} + \text{Join}}{\text{Avg.}} \Rightarrow \text{Sep.} + \text{Join} = 600 \times 10\%$$

$$18 + \text{Join} = 60$$

$$\text{Join} = 42$$

30 Sep.
12 Exp.

Replacement Method - Labour turnover rate = $\frac{\text{No. of replacements}}{\text{Average number of workers}} \times 100$

$$5 = \frac{30}{\text{Average number of workers}} \times 100$$

$$\text{Average number of workers} = 600$$

$$\text{Avg.} = \frac{\text{Op.} + \text{Cl.}}{2}$$

$$600 = \frac{\text{Op.} + \text{Op.} + 24}{2}$$

$$1200 = 2(\text{Op.}) + 24$$

$$\text{Op.} = \frac{1176}{2} = 588$$

Separation Method - Labour turnover rate = $\frac{\text{No. of separations}}{\text{Average number of workers}} \times 100$

$$3 = \frac{\text{No. of separations}}{600} \times 100$$

Number of separations (left and discharged) = 18

Flux Method - Labour turnover rate = $\frac{\text{No. of separations} + \text{No. of recruitments \& joinee}}{\text{Average number of workers}} \times 100$

$$10 = \frac{18 + \text{No. of recruitments \& joinee}}{600} \times 100$$

Number of workers recruited & joined = $60 - 18 = 42$

(a) number of workers recruited and joined = 42

(b) number of workers left and discharged = 18

(c) **Equivalent Employee turnover rate**

Flux Method – Labour turnover rate = $\frac{16}{3} \times 12 = 64\%$

Replacement Method – Labour turnover rate = $\frac{8}{3} \times 12 = 32\%$

Separation Method – Labour turnover rate = $\frac{5}{3} \times 12 = 20\%$

(d) Average workers = $\frac{\text{Opening} + \text{Closing}}{2}$

$$600 = \frac{\text{Opening} + \text{Closing}}{2}$$

Closing = $1200 - \text{Opening}$

Also, Closing workers = Opening workers + recruited & joined – left & discharged

$$1200 - \text{Opening} = \text{Opening} + 42 - 18$$

Opening workers = 588

Question – 2

SK Ltd. is engaged in BPO industry. One of its trainee executives in the Personnel department has calculated labour turnover rate 24.92% for the last year using Flux method.

Following is the data provided by the Personnel department for the last year:

Employees	At the beginning	Joined	Left	At the end	Rep.	Exp.
Data processor	540	1,080	60	1,560	60	1020
Payroll Processors	$40 + 60 - 20 = 80$	20	60	40	20	—
Supervisors	$\frac{20}{2} \times 3 = 30$	60	—	$30 + 60 - 0 = 90$	—	60
Voice Agents	30	20	20	$30 + 20 - 20 = 30$	20	—
Assistant Managers	$30 + 10 - 20 = 20$	20	10	30	10	10
Senior Voice Agents	4	8	—	12	—	8
Senior Data Processors	8	26	—	34	—	26
Team Leaders	$0 + 60 - 0 = 60$	—	60	—	—	—
Total	772	1234	210	1796	110	1124

$$\text{AM} = \frac{2}{3} \times (\text{Sup.} + \text{VA} + \text{AM})$$

$$20 = \frac{2}{3} \times \text{Total}$$

$$\text{Total} = 80$$

Employees transferred from the Subsidiary Company				
Senior Voice Agents	—	8	—	—
Senior Data Processors	—	26	—	—
Employees transferred to the Subsidiary Company				
Team Leaders	—	—	60	—
Assistant Managers	—	—	10	—

At the beginning of the year there were total 772 employees on the payroll of the company. The opening strength of the Supervisors, Voice Agents and Assistant Managers were in the ratio of 3 : 3 : 2.

The company has decided to abandon the post of Team Leaders and consequently all the Team Leaders were transferred to the subsidiary company. The company and its subsidiary are maintaining separate set of books of account and separate Personnel Department.

$$8.57\% = 100 \times \frac{110}{1284}$$

$$\text{Avg. Se} = \frac{772 + 1796}{2} = 1284$$

You are required to calculate:

- (a) Labour Turnover rate using Replacement method and Separation method. $\rightarrow \frac{210}{1284} \times 100 = 16.36\%$
- (b) Verify the Labour turnover rate calculated under Flux method by the trainee executive of the SK Ltd. $\rightarrow \frac{(210 + 1234)}{1284} \times 100 = 112.46\%$

Solution

Working Notes:

- (i) Calculation of no. of employees at the beginning and end of the year

	At the beg.	Joined	Left	At the end	Replacement	Expansion
Data Processors	540	1,080	60	1,560	60	1,020
	[40 + 60 - 20]					
Payroll Proc.	80	20	60	40	20	-
Supervisors*	30	60	-	90	-	60
				[30 + 60 - 0]		
Voice Agents*	30	20	20	30	20	-
Ast. Managers*	20	20	10	30	10	10
Sr. Voice Agents	4	8	-	12	-	8
Sr. Data Processors	8	26	-	34	-	26
Team Leaders	60	-	60	-	-	-
Total	772	1,234	210	1,796	110	1,124

(*) At the beginning of the year:

Strength of Supervisors, Voice Agents and Asst. Managers =

[772 - {540 + 80 + 4 + 8 + 60} employees] or [772 - 692 = 80 employees]

[{Supervisors- $80 \times \frac{3}{8} = 30$, Voice Agents- $80 \times \frac{3}{8} = 30$ & Asst. Managers- $80 \times \frac{2}{8} = 20$ } employees]

At the end of the year:

[Supervisor-(Opening- 30 + 60 Joining) = 90; Voice Agents- (Opening- 30 + 20 Joined – 20 Left) = 30]

(a) Calculation of Labour Turnover:

$$\text{Replacement Method} = \frac{\text{No. of employees replaced during the year}}{\text{Average no. of employees on roll}} \times 100$$

$$= \frac{110}{(772+1,796)/2} \times 100 = 8.57\%$$

$$\text{Separation Method} = \frac{\text{No. of employees separated during the year}}{\text{Average no. of employees on roll}} \times 100$$

$$= \frac{210}{(772+1,796)/2} \times 100 = 16.36\%$$

(b) Labour Turnover under Flux Method

$$\text{Flux Method} = \frac{\text{No. of employees (joined + separated) during the year}}{\text{Average no. of employees on roll}} \times 100$$

$$= \frac{1,234+210}{(772+1,796)/2} \times 100 = 112.46\%$$

Labour Turnover calculated by the executive trainee of the Personnel department is incorrect as it has not taken the No. of new recruitment while calculating the labour turnover under Flux method.

Question – 3

Super Ltd, a manufacturing company is facing the problem of high labour turnover in the factory. Before analyzing the causes and taking remedial steps, the management of the company wants to ascertain the profit lost for the year 2022-23 on account of labour turnover. For this purpose, it has given you the following information:

- (i) Sales for the last year 2022-23 was ₹2,16,80,000 and P/V ratio was 15%. Cont. = 21680000 × 15% = 32.52L → P/L = $\frac{32.52}{4.762} = 6.8319$
- (ii) The total number of actual hours worked by the direct labour force was 5,00,000 hours. The actual direct labour hours included 60,000 hours attributable to training new recruits, out of which 40% of the hours were unproductive. SL - (60000 × 40%) = 4.762
- (iii) Due to delays by the Personnel Department in filling vacancies on account of labour turnover, 95,000 potential productive hours (excluding unproductive training hours) were lost. Total hrs lost = (60000 × 40%) + 95000 = 119000
- (iv) 1,500 units of the output produced during training period were defective. Cost of rectification of defective units was ₹40 per unit. 1500 × 40 = 60000
- (v) Settlement cost of the workers leaving the organization was ₹2,37,880.
- (vi) Recruitment and Selection cost was ₹1,40,000.
- (vii) Cost of Training and Induction was ₹1,61,950.

Assuming that the potential production lost as a consequence of labour turnover could have been sold at prevailing prices, find the profit lost for the year 2022-23 on account of labour turnover.

$$\text{Cont. lost} = 119000 \times 6.8319 = 8,12,996$$

Rect.
Sav.

$$\begin{array}{r} 60000 \\ 539830 \\ \hline 1412826 \end{array}$$

Saving total = 539830

Solution

Computation of productive hours = Actual hours work – Unproductive hours
 $= 5,00,000 - 24,000 = 4,76,000$

Productive hours lost = Potential loss of hours + unproductive training hours
 $= 95,000 + 24,000 = 1,19,000$

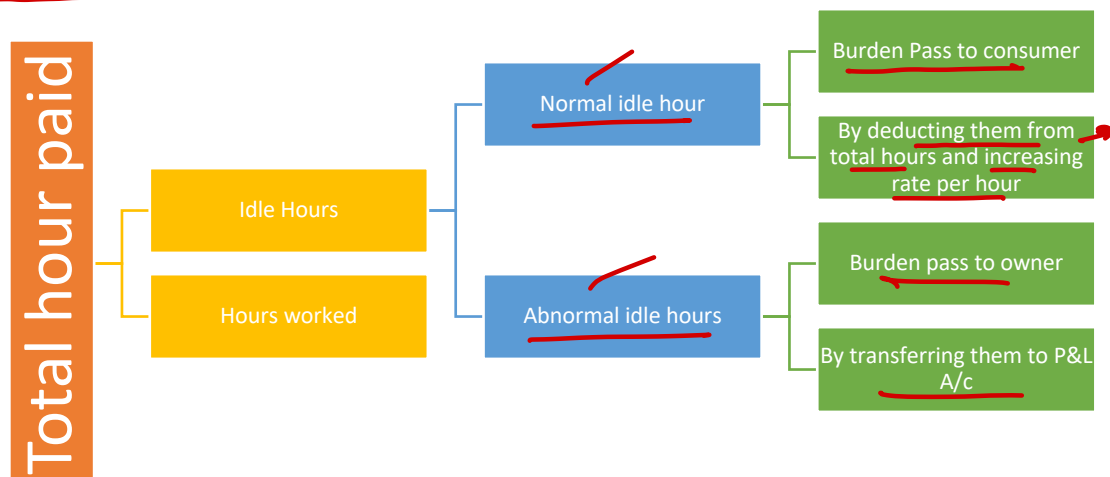
Loss of contribution = $\frac{\text{Total contribution}}{\text{Productive hours}} \times \text{Hours lost}$
 $= \frac{2,16,80,000 \times 15\%}{4,76,000} \times 1,19,000 = ₹8,13,000$

Statement of Profit foregone

Particulars	(₹)
Contribution foregone	8,13,000
Settlement cost due to leaving	2,37,880
Recruitment cost and selection cost	1,40,000
Training and Induction costs	1,61,950
Cost of rectification (1,500 units × ₹40)	60,000
Profit foregone	14,12,830

3. Idle Time

$$\text{Wage Rate} = \frac{\text{Total Wages}}{(\text{Total Hrs.} - \text{No. idle Hrs.})}$$

4. Statement of Wage Rate Calculation

Particulars	Amount (₹)
Basic Wages ✓	--
Dearness Allowance ✓	--
Bonus ✓	--
Commission ✓	--
Perquisite ✓	--
Overtime ✓	--
Any other allowances ✓	--
Employer contribution to PF/ESI etc.	--
Gross Wages / CTC →	--
Effective Working Hours (Total hours – Normal Idle Hours)	--
Wage rate per hour	--

5. Net Wages or In-hand wages calculation

Particulars	Amount (₹)
Gross Wages →	--
(-) Employee contribution to PF/ESI etc.	--
(-) Employer contribution to PF/ESI etc.	--
(-) Tax deducted at source (TDS)	--
(-) Professional Tax	--
(-) Any other deduction	--
Net Wages Payable	--

Question – 4

Following data have been extracted from the books of M/s ABC Private Limited:

Salary (each employee, per month)	₹ 30,000 × 12	=	3,60,000
Bonus	25% of salary × 3.6l	=	90,000
Employer's contribution to PF, ESI etc.	15% of salary × 3.6l	=	54,000
Total cost at employees' welfare activities	₹ 6,61,500 per annum ÷ 175	=	3780
Total leave permitted during the year	→ 30 days		<u>507780</u>
Number of employees	→ 175		
Normal idle time	→ 70 hours per annum		
Abnormal idle time (due to failure of power supply)	→ 50 hours ✓		
Working days per annum	→ 310 days of 8 hours		
			<u>2170</u>

You are required to calculate:

- 1) Annual cost of each employee — 507780
- 2) Employee cost per hour → 507780 ÷ 2170 = ₹.234
- 3) Cost of abnormal idle time, per employee
→ 50 × ₹.234 = ₹.11700

Solution**Calculation of effective hours**

Total working hours (310 × 8)	2,480
Less: Leave days (30 × 8)	<u>240</u>
Available working hours	2,240
Less: Normal loss	<u>70</u>
Effective working hours	<u>2,170</u>

Statement of employee cost per hour

Particulars	Amount (₹)
Salary (30,000 × 12)	3,60,000
Bonus (25% × 3,60,000)	90,000
Employees contribution to PF (15% × 3,60,000)	54,000
Employee welfare (6,61,500 ÷ 175)	3,780
Total Annual Cost (A)	5,07,780
Effective working hours (B)	2,170
Employee cost per hour (A ÷ B)	234

Cost of abnormal idle time per employee = ₹ 234 × 50 hours = ₹ 11,700

$\text{No. Wage Rate} = \text{Rs. 900 Per Hour}$
 $\text{Overtime} = 50\% \text{ Extra}$

$\rightarrow \text{Overtime Rate} = 900 + 50\% = 1350$
 $\text{No.} = 900 \text{ or Premium} = 450$

6. Overtime

- It is hours worked over and above the normal working hours.
- According to Factories Act of 1948, a worker is entitled for overtime at double the rate of his wages if he works more than 9 hours in a day or more than 48 hours in a week.

Overtime payment = Overtime hours × Overtime Rate

7. Overtime Rate & Overtime Premium

- The rate at which overtime hours are paid is known as overtime rate.
- Overtime premium is the extra amount of wages paid over the normal rate.
- Overtime Rate = Normal wage rate + Overtime Premium

Question – 5

A total of 108 labour hours have been put in a particular job card for repair work engaging a semi-skilled and skilled labour (Mr. Deep and Mr. Sam respectively).

The hours devoted by both the workers individually on daily basis for this particular job are given below:

	Monday	Tuesday	Wednesday	Thursday	Friday
→	10.5	8.0	10.5	9.5	10.5
→					

The skilled labour also worked on Saturday for 10 hours.
(Mr. Sam)

Sunday is a weekly holiday and each worker has to work for 8 hours on all week days and 5 hours on Saturdays; the workers are however paid full wages for Saturday (8 hours for 5 hours worked).

Semi-skilled and skilled worker is paid ordinary wage @ ₹ 400 and ₹ 600 respectively per day of 8 hours labour. Further, the workers are also paid dearness allowance @20%. Extra hours worked over and above 8 hours are also paid at ordinary wage rate however, overtime premium of 100% of ordinary wage rate is paid if a worker works for more than 9 hours in a day and 48 hours in a week.

You are required to compute the wages payable to Mr. Deep (semi-skilled) and Mr. Sam (skilled).

	<u>A. Hour</u>	<u>Single</u>	<u>Double</u>	<u>Equiv. Single</u>	<u>A. Hour</u>	<u>Single</u>	<u>Double</u>	<u>Equiv. Single</u>
Mon	10.5	9	1.5	9 + 3 = 12	10.5	9	1.5	9 + 3 = 12
Tu	8	8	-	8 + 0 = 8	8	8	-	8 + 0 = 8
We	10.5	9	1.5	9 + 3 = 12	10.5	9	1.5	9 + 3 = 12
Th	9.5	9	0.5	9 + 1 = 10	9.5	9	0.5	9 + 1 = 10
Fr.	10.5	9	1.5	9 + 3 = 12	10.5	9	1.5	9 + 3 = 12
Sat.	-	-	-	-	10	9	1	9 + 2 = 11
Total	49	44	5	54	59	53	6	65
				@ 60				@ 90
				3240				5850

Solution**Calculation of total normal hours to be paid for Mr. Deep (Semi-skilled)**

Day	Actual hours	Normal hours	Extra Hours	Overtime hours	Equivalent normal hours for overtime worked	Total normal hours payable
	A	B	C	D = A – B	E = D × 2	F = B + C + E
Monday	10.5	8	1	1.5	3	12
Tuesday	8	8	-	-	-	8
Wednesday	10.5	8	1	1.5	3	12
Thursday	9.5	8	1	0.5	1	10
Friday	10.5	8	1	1.5	3	12
Saturday	-	-	-	-	-	-
Total	49	40	4	5	10	54

Calculation of total normal hours to be paid for Mr. Sam (Skilled)

Day	Actual hours	Normal hours	Extra Hours	Overtime hours	Equivalent normal hours for overtime worked	Total normal hours payable
	A	B	C	D = A – B	E = D × 2	F = B + C + E
Monday	10.5	8	1	1.5	3	12
Tuesday	8	8	-	-	-	8
Wednesday	10.5	8	1	1.5	3	12
Thursday	9.5	8	1	0.5	1	10
Friday	10.5	8	1	1.5	3	12
Saturday	10	5	3 + 1 = 4	1	2	11
Total	59	45	8	6	12	65

Note: Mr. Sam will be paid for equivalent 8 normal working hours at ordinary wage rate, though 5 hours of working is required on Saturday because in question it is mentioned that both condition of 9 hour per day and 48 hour a week has to be satisfied. Thus, only 1 hour of overtime over 9 hours will be paid at overtime rate.

Wages Payable

Particulars	Mr. Deep	Mr. Sam
Basic wage per hour	$400 \div 8 = 50$	$600 \div 8 = 75$
Dearness allowance per hour @ 20%	10	15
Hourly wage rate	60	90

Total normal hours payable	54	65
Total wages payable	3,240	5,850

8. Treatment of Overtime Premium

Cause of Overtime	Treatment
Resorted at the desire of the customer	Overtime premium to be charged to the specific customer
Resorted due to general pressure of work to increase the output or irregularity in production	Overtime premium to be charged to general overheads
Resorted due to negligence or delay of a particular department	Overtime premium to be charged to the concerned department
Resorted due to abnormal reasons	Overtime premium to be charged to costing profit & loss account
Resorted due to shortage of labour in industry	Inflate the wage rate and charge it to all customers.

$$\text{Wage Rate} = \frac{3,90,000}{(8-0.5) \times 160 \times 50} = \frac{3,90,000}{60,000} = 6.50$$

Question – 6

In a factory there are 50 workers, working 8 hours per day including 30 minutes for lunch break, worked for 160 days during a period of six months ended on 31st December, 2023. During this period total employee's cost was recorded ₹3,90,000. The management of the factory decided the overtime premium rates for the month of January 2024 as under:

- Sundays and holidays → 180% of basic wages rate → $180\% \times 6.50 = 11.70$
 Before and after normal working hours → 160% of basic wages rate → $160\% \times 6.50 = 10.40$

During the last six months (ended on 31st December, 2023), the following hours were worked:

Normal time	$\text{Avg. Rate} = \frac{4,05,600}{60,000} = 6.76$ <u>Inflated</u>	$6.50 \times 56,250 = 3,65,625$
Sundays and holidays		$11.70 \times 750 = 8,775$
Before and after normal working hours		$10.40 \times 3,000 = 31,200$
Total hours		60,000 <u>4,05,600</u>

During the month of January 2024, the factory worked on job BX in the following manner.

Normal working	✓ 2,400 men hours
Overtime on Sundays and holidays	200 men hours
Overtime before and after normal working	400 men hours
Total hours	<u>3,000</u>

You are required to calculate the labour cost chargeable to job BX and overheads in each of the following situations: $\Delta L = 3,000 \times 6.76 = 20,280$

- Where overtime is worked regularly in whole year as a policy on account of shortage of workers. $\Delta L = 3,000 \times 6.50 = 19,500$ $OH = [200 \times (11.70 - 6.50)] + [400 \times (10.40 - 6.50)] = 2,600$
- Where overtime is worked irregularly to meet the requirement of production.
- Where overtime is worked at the request of the customer to complete the job in time.
- Where overtime is worked on account of flood in the area. $\Delta L = 19,500$ $C.P\&L = 2,600$ $\Delta L = 19,500 + 2,600 = 22,100$

Solution**Workings****Calculation of Basic Wage Rate**

Effective Working Hours for six months ended 31st December, 2023

$$= 50 \text{ workers} \times (8 - 0.5) \text{ hours per day} \times 160 \text{ days} = 60,000 \text{ hours}$$

$$\text{Basic Wage Rate} = \frac{\text{Rs. } 3,90,000}{60,000} = ₹6.50 \text{ per hour}$$

Basic wage rate : ₹6.50 per hour

Overtime wage rate for Sundays and holidays : ₹6.50 × 180% = ₹11.70 per hour

Overtime wage rate before and after working hours : ₹6.50 × 160% = ₹10.40 per hour

Computation of average inflated wage rate (including overtime premium):

Particulars	(₹)
Annual wages for the normal time (56,250 hrs. × ₹6.50)	3,65,625
Wages for overtime on Sundays and holidays (750 hrs. × ₹11.70)	8,775
Wages for overtime before and after working hours (3,000 hrs. × ₹10.40)	31,200
Total wages for 60,000 hrs.	4,05,600

$$\text{Average inflated wage rate} = \frac{\text{Rs. } 4,05,600}{60,000 \text{ hours}} = ₹6.76 \text{ per hour}$$

(i) Where overtime is worked regularly as a policy due to shortage of workers:

The overtime premium is treated as a part of employee cost and job is charged at an inflated wage rate. Hence, employee cost chargeable to job BX

$$= \text{Total hours} \times \text{Inflated wage rate} = 3,000 \text{ hrs.} \times ₹6.76 = ₹20,280$$

(ii) Where overtime is worked irregularly to meet the requirements of production:

Basic wage rate is charged to the job and overtime premium is charged to factory overheads as under:

Employee cost chargeable to Job BX: 3,000 hours @ ₹6.50 per hour = ₹19,500

$$\begin{aligned} \text{Factory overhead} &= \{200 \text{ hrs.} \times ₹(11.70 - 6.50)\} + \{400 \text{ hrs} \times ₹(10.40 - 6.50)\} \\ &= \{₹1,040 + ₹1,560\} = ₹2,600 \end{aligned}$$

(iii) Where overtime is worked at the request of the customer to complete the job in time.

Basic wage rate is charged to the job and overtime premium is charged to job as under:

Employee cost chargeable to Job BX: 3,000 hours @ ₹6.50 per hour = ₹19,500

$$\begin{aligned} \text{Overtime premium} &= \{200 \text{ hrs.} \times ₹(11.70 - 6.50)\} + \{400 \text{ hrs} \times ₹(10.40 - 6.50)\} \\ &= \{₹1,040 + ₹1,560\} = ₹2,600 \end{aligned}$$

$$\text{Total} = ₹19,500 + ₹2,600 = ₹22,100$$

(iv) Where overtime is worked on account of flood in the area

Basic wage rate is charged to the job and overtime premium is charged to costing P & L Account as under:

Employee cost chargeable to Job BX: 3,000 hours @ ₹6.50 per hour = ₹19,500

$$\begin{aligned}\text{Costing P\&L A/c} &= \{200 \text{ hrs} \times ₹(11.70 - 6.50)\} + \{400 \text{ hrs} \times ₹(10.40 - 6.50)\} \\ &= \{₹1,040 + ₹1,560\} = ₹2,600 \\ \text{Total} &= ₹19,500 + ₹2,600 = ₹22,100\end{aligned}$$

9. Wage Payment System

(A) Time Rate System

$$\text{Wages} = \text{No. of hours worked} \times \text{Rate per hour}$$

(B) Piece Rate system

$$\text{Wages} = \text{No. of units produced} \times \text{Rate per unit (or Piece rate)}$$

10. Incentive Plans

(A) Halsey Plan

$$\text{Total Earnings} = (H \times R) + [50\% \times (S - H) \times R]$$

(B) Halsey-wier Plan

$$\text{Total Earnings} = (H \times R) + [33.33\% \times (S - H) \times R]$$

(C) Rowan Plan

$$\text{Total Earnings} = (H \times R) + \left[\left(\frac{S-H}{S} \right) \times H \times R \right]$$

Where, H = Actual hours worked

R = Rate per hour

S = Standard hours or time allowed

$(S - H)$ = Time saved

Actual hrs. = 8 hrs.
Std. hrs. = 10 hrs.
Wage Rate = ₹.120

Halsey

$$\begin{aligned}\text{Wages} &= (8 \times ₹120) + [(10-8) \times 120 \times 50\%] \\ &= ₹.960 + ₹.120 \\ &= ₹.1080\end{aligned}$$

$$\text{Eff. wage Rate} = \frac{1080}{8} = ₹.135$$

11. Effective Hourly Rate of Earning

$$\text{Effective hourly rate of earning} = \frac{\text{Total Wages}}{\text{Actual hours worked}}$$

12. Beneficial Plan

Actual hours = 50% of Standard hours	Halsey Bonus = Rowan Bonus
Actual hours > 50% of Standard hours Or Time Saved < 50% of Standard hours	Halsey Bonus < Rowan Bonus
Actual hours < 50% of Standard hours Or Time Saved > 50% of Standard hours	Halsey Bonus > Rowan Bonus

Question – 7

You are given the following information of a worker:

- (a) Name of worker : → 'S'
 (b) Ticket No. : → 002
 (c) Work started : 1 April at 8am
 (d) Work finished : 5 April at 12 noon

$$SH = 2000 \div 40 = 50 \text{ Hrs.}$$

$$\begin{array}{rcl} 1 \text{ April} & = & 9 \\ 2 \text{ " } & = & 9 \\ 3 \text{ " } & = & 9 \\ 4 \text{ " } & = & 9 \\ 5 \text{ " } & = & 4 \\ \hline \text{AH} & = & 40 \text{ Hrs.} \end{array}$$

- (e) Work allotted : Production of 2,000 units
 (f) Work done and approved : → 2,000 units
 (g) Time and units allowed : → 40 units per hour
 (h) Wage rate : → ₹25 per hour
 (i) Bonus : 40% of time saved
 (j) Worker S worked 9 hours a day

Halsey

$$= (40 \times 25) + [(50 - 40) \times 25 \times 40\%]$$

$$= 1100$$

Rowan

$$= (40 \times 25) + [(50 - 40) \times 25 \times \frac{40}{50}]$$

$$= \text{Rs. } 1200$$

You are required to calculate the remuneration of the worker on the following basis:

- a) Halsey plan and
 b) Rowan plan

Solution

Remuneration under Halsey Plan = $(40 \times 25) + (40\% \times 10 \times 25) = 1,000 + 100 = ₹1,100$

Remuneration under Rowan Plan = $(40 \times 25) + (10 \times 25 \times 40 / 50) = 1,000 + 200 = ₹1,200$

Working note:

Time worked	1.4.11	=	9 hours
	2.4.11	=	9 hours
	3.4.11	=	9 hours
	4.4.11	=	9 hours
	5.4.11	=	<u>4 hours</u>
			<u>40 hours</u>

Time allowed = 50 hours (2,000/40)

Time taken = 40 hours

Time saved = 10 hours

Rowan

$$(37.50)(H) = (H \times 30) + [(4 - H) \times 30 \times \frac{H}{4}]$$

$$(37.50)(H) = 30H + (4 - H)(7.5)(H)$$

$$(37.50)(H) = 30H + 30H - 7.5H^2$$

$$(7.50)(H) = H(60 - 7.5H)$$

$$37.50 = 60 - 7.5H$$

Halsey

$$(3 \times 30) + [(4 - 3) \times 30 \times 50\%] = 105$$

$$\text{Eff wage rate} = \frac{105}{3} = \text{Rs. } 35$$

Question – 8

A skilled worker in SK Ltd. is paid a guaranteed wage rate of ₹ 30 per hour. The standard time per unit for a particular product is 4 hours. S, a machine man, has been paid wages under the Rowan Incentive Plan and he had earned an effective hourly rate of ₹ 37.50 on the manufacture of that particular product. What could have been his total earnings and effective hourly rate, had he been put on Halsey Incentive Scheme (50%)?

Solution

Let actual time taken by the worker S = H

$$\text{Total wages in Rowan plan} = (H \times 30) + \left(\frac{H}{4}\right) \times (4 - H) \times 30$$

$$(H \times 37.50) = 30H + 30H - 7.5H^2$$

$$22.5H = 7.5H^2$$

$$H = 3 \text{ hours}$$

$$\text{Total wages of workman in Halsey scheme} = (3 \times 30) + \frac{50}{100} \times (4 - 3) \times 30 = ₹ 105$$

$$\text{Effective hourly rate of earnings under Halsey Plan} = \frac{105}{3 \text{ hours}} = ₹ 35$$

$$\begin{aligned} & \frac{S}{x} \quad \frac{K}{x} \\ & 30y \quad 40y \\ & (50-30)xy \times \frac{30}{50} = 12y \quad (50-40)xy \times \frac{50}{100} = 5y \\ & S \times 30 = 150 \quad 40 \times 5 = 200 \\ & x + 42y + 150 \quad x + 45y + 200 \\ & x + 42y + 150 = 3490 \\ & x + 45y + 200 = 3600 \\ & -3y - 50 = -110 \\ & \therefore x = 3490 - 150 - (12)/20 \quad y = \frac{60}{3} = 20 \\ & = 2500 \end{aligned}$$

Question – 9

Two workers 'S' and 'K' produce the same product using the same material. Their normal wage rate is also the same. 'S' is paid bonus according to Rowan scheme while 'K' is paid bonus according to Halsey scheme. The time allowed to make the product is 50 hours. 'S' takes 30 hours while 'K' takes 40 hours to complete the product. The factory overhead rate is ₹ 5 per person-hour actually worked. The factory cost of product manufactured by 'S' is ₹ 3,490 and for product manufactured by 'K' is ₹ 3,600.

Required:

- Compute the normal rate of wages
- Compute the material cost
- Prepare a statement comparing the factory cost of the product as made by two workers.

Solution

Let x be the cost of material and y be the normal rate of wages per hour

Statement of Factory Cost

Particulars	Worker S	Worker K
Material	₹ x	₹ x
Wages	30y	40y

Bonus (A = $30y \times 20/50$) (B = $10y \times 50\%$)	12y	5y
Overheads @ ₹ 5 per person hour worked	150	200
	$x + 42y + 150$	$x + 45y + 200$

The following two equations can be made

$$x + 42y + 150 = ₹ 3,490 \quad \dots(i)$$

$$x + 45y + 200 = ₹ 3,600 \quad \dots(ii)$$

On subtracting equation (i) from equation (ii)

$$3y + 50 = 110$$

$$\text{or } 3y = 110 - 50$$

$$y = 60/3 = 20$$

On substituting the value of y in equation (i)

$$x + 840 + 150 = 3,490$$

$$\text{or } x = 3,490 - 990$$

$$\text{or } x = 2,500$$

Thus:

(a) Normal Wage Rate is ₹ 20 per hour

(b) Cost of material used for the product is ₹ 2,500

(c) Statement of Cost

Particulars	Worker S	Worker K
Material	2,500	2,500
Wages	600	800
Bonus	240	100
Overheads @ ₹ 5 per person hour worked	150	200
	3,490	3,600

Question – 10

Mr. S is working by employing 10 skilled workers. He is considering the introduction of some incentive scheme - either Halsey Scheme (with 50% bonus) or Rowan Scheme - of wage payment for increasing the labour productivity to cope with the increased demand for the product by 25%. He feels that if the proposed incentive scheme could bring about an average 20% increase over the present earnings of the workers, it could act as sufficient incentive for them to produce more and he has accordingly given this assurance to the workers.

$$\text{Halsey} \rightarrow (2000 \times 2) + [500 \times 2 \times 50\%] = 4500 \div 2000 = 2.25 \checkmark$$

$$\text{Rowan} \rightarrow (2000 \times 2) + [500 \times 2 \times \frac{2000}{2500}] = 4800 \div 2000 = 2.40 \checkmark$$

As a result of this assurance, the increase in productivity has been observed as revealed by the following figures for the current month:

$$\text{Std. Cost Per Piece} = 2 \text{ hrs.} \times ₹.2 = ₹.4$$

$$\text{Halsey} = \frac{4500}{1250} = 3.60 \rightarrow \text{Sav.} = 4 - 3.60 = 0.40$$

$$\text{Rowan} = \frac{4800}{1250} = 3.84 \rightarrow \text{Sav.} = 4 - 3.84 = 0.16$$

Cost & Management Accounting

$$\text{Halsey} \rightarrow \frac{(2.25 - 2) \times 100}{2} = 12.5\%$$

$$\text{Rowan} \rightarrow \frac{(2.40 - 2) \times 100}{2} = 20\% \rightarrow \text{Select due to Assurance}$$

$$SH = 1250 \times 2 = 2500 \quad AH = 25 \times 8 \times 10 = 2000 \quad R = 2 \quad TS = 2500 - 2000 = 500$$

Hourly rate of wages (guaranteed)

→ ₹ 2.00 (R)

Average time for producing 1 piece by one worker at the previous performance

→ 2 hours

(This may be taken as time allowed)

No. of working day in the month

→ 25

No. of working hours per day for each worker

→ 8

Actual production during the month

→ 1,250 units

Required:

- Calculate effective rate of earnings per hour under Halsey Scheme and Rowan Scheme.
- Calculate the savings to Mr. S in terms of direct labour cost per piece under the above schemes.
- Advice Mr. S about the selection of the scheme to fulfill his assurance.

Solution

Actual hours = $25 \times 8 \times 10 = 2000$; Standard hours = $1,250 \times 2 = 2,500$; Wage rate = 2

$$(a) \text{ Earning under Halsey scheme} = (2,000 \times 2) + \frac{50}{100} \times (2,500 - 2,000) \times 2 = ₹ 4,500$$

$$\text{Effective hourly rate of earnings under Halsey Plan} = \frac{4,500}{2,000 \text{ hours}} = ₹ 2.25$$

$$\text{Earnings under Rowan plan} = (2,000 \times 2) + \left(\frac{2,000}{2,500} \right) \times (2,500 - 2,000) \times 2 = ₹ 4,800$$

$$\text{Effective hourly rate of earnings under Rowan Plan} = \frac{4,800}{2,000 \text{ hours}} = ₹ 2.40$$

$$(b) \text{ Labour cost per piece under time wage system} = 2 \times 2 = ₹ 4$$

$$\text{Labour cost per piece under Halsey} = \frac{4,500}{1,250} = ₹ 3.60$$

$$\text{Savings per piece under Halsey Scheme} = 4 - 3.60 = ₹ 0.40$$

$$\text{Labour cost per piece under Rowan} = \frac{4,800}{1,250} = ₹ 3.84$$

$$\text{Savings per piece under Rowan Scheme} = 4 - 3.84 = ₹ 0.16$$

- As per above, it is better for Mr. S to adopt Halsey Scheme but since he has assured workers of an average 20% increase over the present earnings, he will have to select Rowan Scheme as is evident from the following:

$$\text{Increase in earning under Halsey Scheme} = \frac{4,500 - 4,000}{4,000} \times 100 = 12.5\%$$

$$\text{Increase in earning under Rowan Scheme} = \frac{4,800 - 4,000}{4,000} \times 100 = 20\%$$

Question – 11

A Company is undecided as to what kind of wage scheme should be introduced. The following particulars have been compiled in respect of three workers. Which are under consideration of the management.

	I	II	III
Actual hours worked	380	100	540
Hourly rate of wages (in ₹)	40	50	60
Production in units:			
- Product S @15	210	-	600
- Product K @20	360	-	1350
- Product M @30	460	250	-
Standard time allowed per unit of each product is:			
	S	K	M
Minutes	15	20	30

For the purpose of piece rate, each minute is valued at ₹ 1/- $15 \times 1 = 15$ $20 \times 1 = 20$ $30 \times 1 = 30$

You are required to calculate the wages of each worker under:

- Guaranteed hourly rate basis →
- Piece rate earning basis, but guaranteed at 75% of basic pay (Guaranteed hourly rate if his earnings are less than 50% of basic pay).
- Premium bonus basis where the worker received bonus based on Rowan scheme.

Solution**(a) Computation of wages of each worker under guaranteed hourly rate basis**

Worker	Actual hours worked	Hourly wage rate	Wages (₹)
I	380 ✓	✗ ✓ 40	= 15,200
II	100	✗ 50	= 5,000
III	540	✗ 60	= 32,400

(b) Computation of wages of each worker under piece work earning basis

Product	Piece rate per unit	Worker-I		Worker-II		Worker-III	
		Units	Wages	Units	Wages	Units	Wages
S	✓ 15	- 210	3,150	- -	-	600	9,000
K	✓ 20	- 360	7,200	-	-	1,350	27,000
M	✓ 30	- 460	13,800	→ 250	7,500	-	-
Total			24,150		7,500		36,000

Since each worker's earnings are more than 50% of basic pay. Therefore, worker-I, II and III will be paid the wages as computed i.e. ₹ 24,150, ₹ 7,500 and ₹ 36,000 respectively.

(c) Computation of wages of each worker under Rowan scheme

Worker	Time Allowed <u>W.A.</u>	Time Taken	Time Saved ✓	Wage rate per hour	Earnings	Bonus	Total Earning
I	402.5	✓ 380	22.5	40	15,00	850	16,050
II	125	✓ 100	25	50	5,000	1,000	6,000
III	600	✓ 540	60	60	32,400	3,240	35,640

Working Notes:

(1) Piece rate per unit

Product	Standard time per unit in minute	Piece rate per minute	Piece rate per unit
S	15	1	15
K	20	1	20
M	30	1	30

(2) Time allowed to each worker

Worker	Product S	Product K	Product M	Total hours
I	$210 \times 15 = 3,150$	$360 \times 20 = 7,200$	$460 \times 30 = 13,800$	$24,150 \div 60 = 402.5$ ✓
II	-	-	$250 \times 30 = 7,500$	$7,500 \div 60 = 125$ ✓
III	$600 \times 15 = 9,000$	$1,350 \times 20 = 27,000$	-	$36,000 \div 60 = 600$ ✓

- (3) Bonus of worker -I under Rowan = $\frac{380}{402.50} \times 22.5 \times 40 = 850$
 Bonus of worker -II under Rowan = $\frac{100}{125} \times 25 \times 50 = 1,000$
 Bonus of worker -III under Rowan = $\frac{540}{600} \times 60 \times 60 = 3,240$

Question – 12

SK Ltd. operates a boutique which works for various fashion houses and retail stores. It has employed 26 workers and pays them on time rate basis. On an average an employee is allowed 8 hours for boutique work on a piece of garment. In the month of May 2021, two workers S and K were given 15 pieces and 21 pieces of garments respectively for boutique work. The following are the details of their work:

	S	K
Work Assigned	15 pieces	21 pieces
Time Taken	100 hours	140 hours

SH
TS

$$15 \times 8 = 120$$

$$120 - 100 = 20$$

$$21 \times 8 = 168$$

$$168 - 140 = 28$$

$$\text{SH } S = (100 \times 12) + [20 \times 12 \times 50\%] = 1320$$

$$K = (140 \times 12) + [28 \times 12 \times 50\%] = 1848$$

$$\text{TS } S = (100 \times 12) + [20 \times 12 \times \frac{100}{120}] = 1400$$

$$K = (140 \times 12) + [28 \times 12 \times \frac{140}{168}] = 1960$$

Cost & Management Accounting

$$\text{(iii) } S \text{ Loss} = 1400 - 1320 = 80$$

$$K \text{ Loss} = 1960 - 1848 = 112$$

Workers are paid bonus as per Halsey System. The existing rate of wages is ₹ 60 per hour. As per the new wages agreement the workers will be paid ₹ 72 per hour w.e.f. 1st June 2021. At the end of the month May 2021, the accountant of the company has wrongly calculated wages to these two workers taking ₹ 72 per hour.

Required:

- Calculate the loss incurred due to incorrect rate selection.
- Calculate the loss incurred due to incorrect rate selection, had Rowan scheme of bonus payment followed.
- Calculate the loss/savings if Rowan scheme of bonus payment had followed.

Solution

Basic Calculation

Particulars	S	K
Actual hours taken	100 hrs.	140 hrs.
Standard hours	$15 \times 8 = 120$ hrs.	$21 \times 8 = 168$ hrs.
Hours saved	20 hrs.	28 hrs.

(i) Statement of calculation of loss due to incorrect rate selection

Particulars	S	K
Wages due @ ₹ 60 (A)	$(100 \times 60) + (20 \times 60 \times 50\%) = 6,600$	$(140 \times 60) + (28 \times 60 \times 50\%) = 9,240$
Wages paid @ ₹ 72 (B)	$(100 \times 72) + (20 \times 72 \times 50\%) = 7,920$	$(140 \times 72) + (28 \times 72 \times 50\%) = 11,088$
Extra wages paid (B – A)	1,320	1,848

(ii) Statement of calculation of loss due to incorrect rate selection under Rowan system

Particulars	S	K
Wages due @ ₹ 60 (A)	$(100 \times 60) + (20 \times 60 \times \frac{100}{120}) = 7,000$	$(140 \times 60) + (28 \times 60 \times \frac{140}{168}) = 9,800$
Wages paid @ ₹ 72 (B)	$(100 \times 72) + (20 \times 72 \times \frac{100}{120}) = 8,400$	$(140 \times 72) + (28 \times 72 \times \frac{140}{168}) = 11,760$
Extra wages paid (B – A)	1,400	1,960

(iii) Statement of calculation of saving due to Rowan system

Particulars	S	K
Extra wages paid under Halsey	1,320	1,848
Extra wages paid under Rowan	1,400	1,960
Difference (loss)	(80)	(112)

Question – 13

Rivaan Industries manufactures and sells wooden study tables. The demand for these tables has been steadily increasing due to the company's strong distribution network and reputation for durable products. Despite the rising market potential, the company continues to rely on its long-established wage practices. At present, Rivaan Industries pays wages under a piece-rate system, with a standard piece rate of ₹18 per table produced.

The company policy for payment under the piece-rate system is as follows:

- For efficiency less than 100%:
Workers are paid 80% of the standard piece rate $\rightarrow 80\% \times 18 = 14.40$ ✓
- For efficiency equal to or more than 100%:
Workers are paid 125% of the standard piece rate $\rightarrow 125\% \times 18 = 22.50$ ✓

hrs. per week = $(9 \times 5) + (4 \times 1) = 49$

The standard output per hour is 3 tables. Each worker works 9 hours per day from Monday to Friday and 4 hours on Saturday. A total of 132 workers are currently eligible for this wage plan.

However, the cutting and finishing machines used in production were purchased many years ago and now limit workers' efficiency. As a result, workers have demanded a shift to a time-rate wage system at ₹65 per hour, along with incentives under the Halsey Premium Plan (50% sharing of time saved).

To evaluate the financial impact, the management has prepared the following estimated production levels for January 2026 under three different scenarios:

$AM = 49 \times 4 \times 132 = 25872$ 25872 25872

Production Estimates (January 2026)

Scenario	Worst Case	Optimal Case	Best Case
Production (units)	36,900 @ 14.40	73,800 @ 14.40	1,10,700 @ 22.50
SH Eff. (SH/AM × 100)	$36900 \div 3 = 12300$ 47.54%	$73800 \div 3 = 24600$ 95.08%	$110700 \div 3 = 36900$ 142.63%

In addition, the company is assessing a potential Technology Upgrade Case involving partial automation, which may influence both output levels and worker productivity.

Required:

- (a) Compute the Total wages, and Average wages per worker per month, for all three scenarios (Worst, Optimal, Best Case) under each of the following wage systems:

- Current piece-rate wage system (based on differential piece rates)
- Workers' proposed system: Time-rate wages @ ₹65 per hour + Halsey Premium (50%)

- (b) Mr. R, one of the workers, produced 900 tables in December 2025. If he produces the same number of tables in January 2026, compute:

- His earnings under the current piece-rate system $\rightarrow 900 \times \text{Rs. } 22.50 = 20250$
- His earnings under the proposed time-rate + Halsey incentive system $\rightarrow (196 \times 65) + [(300 - 196) \times 65 \times 50\%]$
= 16120
- The difference in his earnings $\rightarrow \text{Loss} = 20250 - 16120 = 4130$

Based on your computation, should Mr. R support the workers' demand?

(Assume 4 working weeks in a month.)

Solution

(a) Calculation of Total wages and average wages per worker per month.

(i) When Current system of wages and incentive payment system is followed:

Standard hours per Month = (9 hrs x 4 days + 4 hrs) x 4 = 196 hours

Standard production per worker = 196 hours x 3 units = 588 units

Standard production for 132 workers = 588 x 132 = 77,616 units

Particulars	Worst Case	Optimal Case	Best Case
I. Standard Production (units) (588 units x 132 workers)	77,616	77,616	77,616
II. Actual Production (units)	✓ 36,900	73,800	1,10,700
III. Efficiency (%) (II ÷ I × 100)	47.54%	95.08%	142.63%
IV. Applicable Piece Rate	₹ 14.40 (₹ 18 x 0.8)	₹ 14.40 (₹ 18 x 0.8)	₹ 22.50 (₹ 18 x 1.25)
V. Total Wages (II × IV)	₹ 5,31,360	₹ 10,62,720	₹ 24,90,750
VI. Average Wage per Worker (V ÷ 132)	₹ 4,025.45	₹ 8,050.91	₹ 18,869.32

(ii) When workers' demand for time rate wages and Halsey premium plan is accepted:

Particulars	Worst Case	Optimal Case	Best Case
I. Actual Production (units)	36,900	73,800	1,10,700
II. Standard Units per hour	3	3	3
III. Standard Hours Required (I ÷ II)	12,300	24,600	36,900
IV. Actual working hours (132 workers × 196 hrs)	25,872	25,872	25,872
V. Time Saved (III - IV)	—	—	11,028 ✓
VI. Time Wages (IV × ₹ 65)	₹ 16,81,680	₹ 16,81,680	₹ 16,81,680
VII. Halsey Bonus (50% × Time Saved × 65)	—	—	₹ 3,58,410 ✓
VIII. Total Wages (VI + VII)	₹ 16,81,680	₹ 16,81,680	₹ 20,40,090
IX. Average Wage per Worker (VIII ÷ 132)	₹ 12,740	₹ 12,740	₹ 15,455.23

(b) Calculation of gain or loss in the current monthly income of Mr. R:

1. Under Current Piece-Rate System

Standard production = 588 units

Production in December = 900 units

Efficiency = $900 \div 588 \times 100 = 153.06\%$

Rate = ₹ 22.50

Total wages = $900 \times 22.50 = ₹ 20,250$

2. Under Proposed Time Rate + Halsey Plan

Standard hours = $900 \text{ units} \div 3 \text{ units} = 300 \text{ hours}$

Actual hours available = 196 hours

Time saved = $300 - 196 = 104 \text{ hrs}$, so bonus = 0

Time wages = $196 \times ₹ 65 = ₹ 12,740$

Halsey Bonus ($50\% \times \text{Time Saved} \times 65$) = $104 \text{ hrs} \times 50\% \times 65 = ₹ 3,380$

Total expected wages = ₹ 16,120

3. Loss To Mr. R

Loss = $20,250 - 16,120 = ₹ 4,130$

Supporting the demand of colleague workers will cost ₹ 4,130 in the next month to Mr. R.

